

RASPBERRY PI NAS – USB 3.x TO NVMe BRIDGE

Board target: 90 mm x 90 mm, 4 layers. SCHEMATIC ONLY – NO PCB LAYOUT.

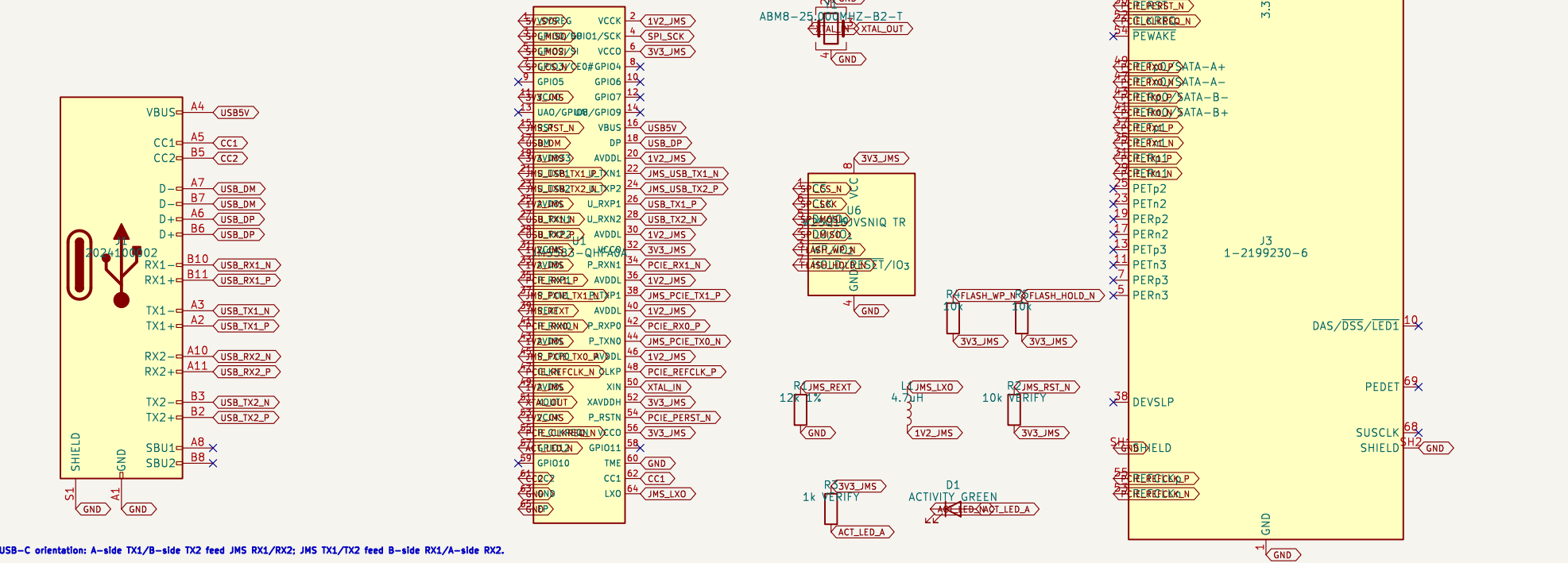
1. USB INTERFACE

2. JMS583 BRIDGE

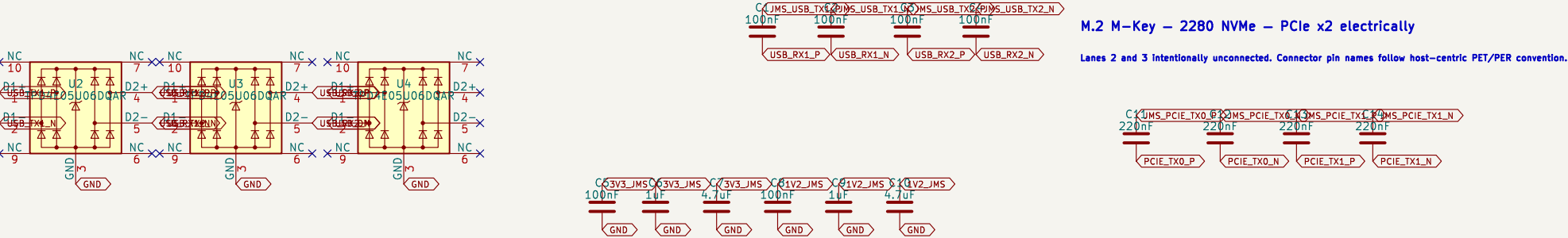
3. NVME / M.2

VERIFY AGAINST JMS583 REFERENCE DESIGN BEFORE PCB RELEASE

JMS583 public pin table used. Confirm EP grounding, reset pull-up, crystal load network, internal-regulator rail topology and LED polarity.



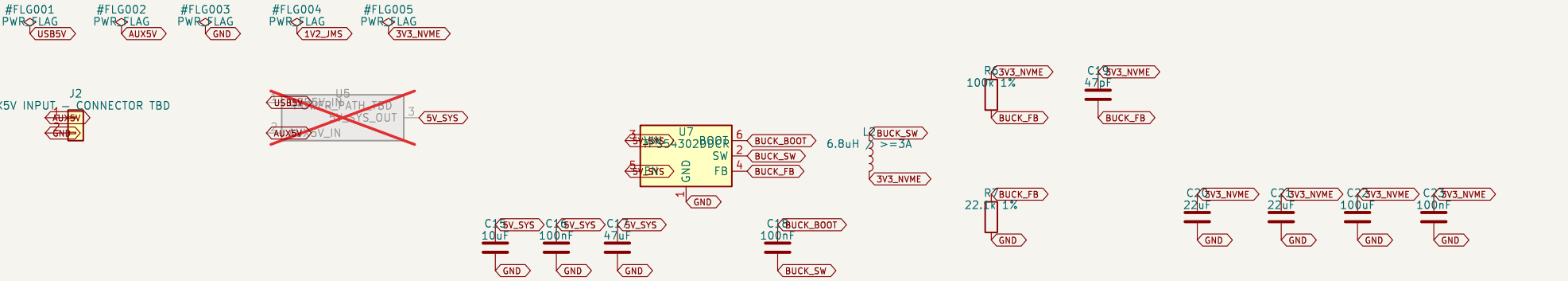
USB-C orientation: A-side TX1/B-side TX2 feed JMS RX1/RX2; JMS TX1/TX2 feed B-side RX1/A-side RX2.



4. POWER

POWER-PATH PLACEHOLDER – NO USB5V/AUX5V SHORT IS PRESENT

Select a reverse-current-blocking 5 V power mux / Ideal-diode solution and rated AUX connector before PCB release.



TP554302: 6.8 uH, 44 uF output, 100k/22.1k divider and 47 pF feed-forward follow TI's 3.3 V table.

Inductor/capacitor voltage, ripple-current, DC-bias and thermal ratings require final MPN selection.